Horizons and Becoming: what an event's past settles, how its standpoint reasons, and where grounding matters

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Abstract

A companion paper shows that, under a locality principle, what has become for an event in a globally hyperbolic spacetime lies between its chronological and its causal past. We ask what this fixes and what it leaves open, and find that the answers are governed by horizon structure. First, being settled from an event (fixed, under deterministic laws, by what is real from it) differs from being real from it; the settled region is essentially Geroch's domain of prediction. We characterize it by a condition on Cauchy slices and show that it can jump: in a closed dust universe nothing outside an event's causal past is settled from it while the universe expands, and everything is settled from every event after maximum expansion. Second, the modal logic of the standpoint frame validates S4 but, at a given standpoint, validates the confluence axiom .2 exactly when any two events in its past have a common past. In big-bang cosmologies .2 therefore fails at every standpoint; the cosmological horizon problem is this failure for our own standpoint, and inflation repairs it only for the observed pairs. Third, if the standpoints that exist are the events on matter worldlines, regions beyond every such worldline's event horizon are real from no standpoint and not fixed by anything real from one. There, and only there, an eternalist and a standpoint-first ontology disagree about what facts there are. For a lone observer in de Sitter spacetime this region is half the universe.

1 Introduction

Stein's theorem says that in Minkowski spacetime a transitive, symmetry-invariant relation of becoming that makes some past event definite is the causal order or trivial [4, 17, 18]. A companion paper [19] extends this to general relativity, replacing symmetry by local covariance: for any non-universal locally covariant becoming theory, $I^-_{\mathrm{s}}(b) \subseteq F(b) \subseteq J^-(b)$ in every globally hyperbolic spacetime, where $F(b)$ is what has become for b; with the principle that whatever can signal to b has become for it, $F(b) = J^-(b)$.

That result fixes the content of an event's "fragment" of reality. It leaves three questions open, each of which a philosopher of time will press.

1. If the future is not real from b, is it at least open from b, or can deterministic laws settle it from what is real?

2. What is the logic of reasoning *from* a standpoint about other standpoints real from it?

3. The view that reality is standpoint-relative agrees with eternalism on every tenseless truth. Is there anywhere the two disagree about what facts there are?

The answers turn on horizons: on whether past cones wrap round space, on whether two events in a past cone share a common past, and on whether some regions lie in the past of no matter at all.

We work throughout with smooth, connected, time-oriented, globally hyperbolic spacetimes (M, g). $I_{\mathrm{s}}^-(p)$ and $I_{\mathrm{s}}^+(p)$ are the strict chronological past and future (via timelike curves), $J^{\pm}(p)$ the causal past and future, $x \leq y$ means $x \in J^-(y)$, and $x \ll y$ means $x \in I_{\mathrm{s}}^-(y)$. We use the standard facts that $\leq$ is closed in $M \times M$, that $J^-(x) \cap S$ is compact for every Cauchy surface S, and push-up: if $y \leq z$ and $z \ll b$ then $y \ll b$ [13]. By Bernal and Sánchez [2] there is a Cauchy temporal function t; we write $S_c = t^{-1}(c)$.

2 Settled versus real

Definition 1. An event x is *settled from* b if every past-inextendible causal curve from x meets $I_{\mathrm{s}}^-(b)$. Write $\mathrm{Set}(b)$ for the set of such events.

If the fields obey deterministic laws with the domain-of-dependence property (data on a region fix the fields at every event all of whose past-inextendible causal curves meet that region), then what happens at a settled event is fixed by what happens in $I_{\mathrm{s}}^-(b)$, which is real from b. Trivially $I_{\mathrm{s}}^-(b) \subseteq \mathrm{Set}(b)$. The notion is not new. Geroch's *domain of prediction* $P(q)$ is the set of x such that every past-directed timelike curve from x without past endpoint enters $I^-(q)$, but $I^-(x) \not\subseteq I^-(q)$ [6, sec. 3]; so, up to the use of causal rather than timelike curves, $\mathrm{Set}(b)$ is $I_{\mathrm{s}}^-(b) \cup P(b)$. What is new here is the reading: $\mathrm{Set}(b)$ is what is settled from b, as opposed to what is real from b. The definition uses the open set $I_{\mathrm{s}}^-(b)$ rather than $J^-(b)$; points of the null cone of b are then not settled from b in Minkowski spacetime (Proposition 5), matching the light-cone freedom of the companion theorem.

Theorem 2. *$x \in \mathrm{Set}(b)$ if and only if there is $c < t(x)$ with $J^-(x) \cap S_c \subseteq I_{\mathrm{s}}^-(b)$.*

Proof. ($\Leftarrow$) A past-inextendible causal curve from x crosses the Cauchy surface S_c, at a point of $J^-(x) \cap S_c \subseteq I_{\mathrm{s}}^-(b)$.

($\Rightarrow$) Suppose that for every $c < t(x)$ the compact set $K_c = (J^-(x) \cap S_c) \setminus I_{\mathrm{s}}^-(b)$ is nonempty (it is compact because $I_{\mathrm{s}}^-(b)$ is open). Fix $c_1 > c_2 > \cdots \to -\infty$ with $c_1 < t(x)$. For each n let

$$P_n = \{(z_1, \ldots, z_n) \in K_{c_1} \times \cdots \times K_{c_n} : x \geq z_1 \geq z_2 \geq \cdots \geq z_n\}.$$

P_n is compact, since $\leq$ is closed. It is nonempty: pick $y \in K_{c_n}$ and a causal curve from x to y; it crosses each S_{c_i}, $i < n$, at a point z_i with $z_i \leq x$, and $z_i \notin I_{\mathrm{s}}^-(b)$, for otherwise $y \leq z_i \ll b$ would give $y \ll b$ by push-up. The projections $P_{n+1} \to P_n$ forgetting the last coordinate are continuous, so by the standard compactness argument (an inverse limit of nonempty compact Hausdorff spaces is nonempty) there is an infinite sequence $x \geq z_1 \geq z_2 \geq \cdots$ with $z_n \in K_{c_n}$. Join consecutive terms by causal curves and concatenate. Each segment avoids $I_{\mathrm{s}}^-(b)$ by push-up as before, and $t \to -\infty$ along the concatenation, so it has no past endpoint: it is a past-inextendible causal curve from x that misses $I_{\mathrm{s}}^-(b)$. Hence $x \notin \mathrm{Set}(b)$. □

Corollary 3. *If $I_{\mathrm{s}}^-(b)$ contains a Cauchy surface, then $\mathrm{Set}(b) = M$.*

Proof. If $S_c \subseteq I_{\mathrm{s}}^-(b)$, then every x with $t(x) > c$ satisfies the condition of Theorem 2, and every x with $t(x) \leq c$ lies in $I_{\mathrm{s}}^-(b)$, since $I_{\mathrm{s}}^-(b)$ is a past set: a point to the past of S_c is in the past of some point of S_c. □

Settledness is defined by causal curves alone, so it is invariant under conformal rescaling of the metric. Conformal diagrams decide it.

Lemma 4. *Let (Σ, d) be Euclidean space $\mathbb{R}^3$, or the unit round sphere S^3 with $r_2 \leq \pi$. Then the closed ball $\bar{B}(X, r_1)$ lies in the open ball $B(Y, r_2)$ if and only if $d(X, Y) + r_1 < r_2$.*

Proof. ($\Leftarrow$) is the triangle inequality. ($\Rightarrow$) Suppose $d(X, Y) + r_1 \geq r_2$. If $d(X, Y) + r_1 \leq \pi$ (always so in $\mathbb{R}^3$), follow a minimizing geodesic from Y through X (any geodesic from X if $X = Y$) a further distance r_1 beyond X; the endpoint is in $\bar{B}(X, r_1)$ at distance $d(X, Y) + r_1 \geq r_2$ from Y. If $d(X, Y) + r_1 > \pi$ on S^3, the antipode of Y is at distance $\pi - d(X, Y) < r_1$ from X and at distance $\pi \geq r_2$ from Y. □

Proposition 5. *Let M be conformal to $\{(\eta, X) : \eta \in (\eta_-, \eta_+),\ X \in \Sigma\}$ with metric $-d\eta^2 + d\Sigma^2$, where Σ is $\mathbb{R}^3$, or the unit S^3 with $\eta_+ - \eta_- \leq \pi$. Then $\mathrm{Set}(b) = I_{\mathrm{s}}^-(b)$ for every b.*

Proof. Write $x = (\eta_x, X)$, $b = (\eta_b, Y)$. Here $x \ll b$ iff $d(X, Y) < \eta_b - \eta_x$ (on S^3 the bound $\eta_+ - \eta_- \leq \pi$ keeps $\eta_b - \eta_x < \pi$). For $\eta_- < c < \eta_x$, $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ and $I_{\mathrm{s}}^-(b) \cap S_c = B(Y, \eta_b - c)$, with $\eta_b - c < \pi$ on S^3. By Lemma 4 the inclusion holds iff $d(X, Y) + \eta_x - c < \eta_b - c$, i.e. iff $x \ll b$. Apply Theorem 2. □

This covers Minkowski spacetime, spatially flat FLRW universes with a big bang (conformal to $\eta \in (0, \eta_+)$), the closed radiation universe $a(\eta) = a_0 \sin \eta$, $\eta \in (0, \pi)$, and global de Sitter spacetime (conformal to $\eta \in (-\pi/2, \pi/2)$ with $\Sigma = S^3$) [10]. In all of them, nothing outside an event's chronological past is ever settled from it.

Proposition 6 (The future closes at maximum expansion)**.** *Let M be conformal to $(\eta_-, \eta_+) \times S^3$ with $\eta_+ - \eta_- > \pi$. Then $\mathrm{Set}(b) = I_{\mathrm{s}}^-(b)$ if $\eta_b \leq \eta_- + \pi$, and $\mathrm{Set}(b) = M$ if $\eta_b > \eta_- + \pi$. In particular, in the closed dust FLRW universe, $a(\eta) = \frac{1}{2} a_m (1 - \cos \eta)$, $\eta \in (0, 2\pi)$, the transition is at $\eta = \pi$, the moment of maximum expansion. In the Einstein static universe ($\eta_\pm = \pm\infty$), $\mathrm{Set}(b) = M$ for every b.*

Proof. If $\eta_b > \eta_- + \pi$, choose $c \in (\eta_-, \eta_b - \pi)$. Then $I_{\mathrm{s}}^-(b) \cap S_c$ is the open ball of radius $\eta_b - c > \pi$, i.e. all of S_c, and Corollary 3 applies. If $\eta_b \leq \eta_- + \pi$, every admissible $c > \eta_-$ has $\eta_b - c < \pi$, and the argument of Proposition 5 applies verbatim. □

Remark 7. Under deterministic laws, then, the closed dust universe has a sharp *closing of the future*. While it expands, what is real from an event (its past) settles nothing beyond itself. From every event after maximum expansion, the entire future, indeed the entire spacetime, is settled. The closed radiation universe and de Sitter spacetime never close; the Einstein static universe is closed from every event. Whether "the future is open" is thus not a question for logic or for temporal ontology alone; given the laws, it is a question about the global causal structure, and it can have different answers at different cosmic epochs.

Relation to prior work. Geroch gives an equivalent form of his definition, by an achronal surface $S \subseteq I^-(q)$ with $x \in D^+(S)$ [6, sec. 3]; Theorem 2 is a slicing version of this for globally hyperbolic spacetimes. Geroch already notes the Einstein static universe, where prediction reaches everything [6, Fig. 5], and conjectured that prediction of an event to the future of q requires a compact spacelike slice; that conjecture was refined and proved by Manchak [12] (see also [15]). Proposition 6 is consistent with it: the future closes only in spatially closed universes. We have not found the dust threshold at maximum expansion, or the contrast between settled and real, in the sources read. Manchak also shows that an observer who does not already know the spacetime cannot verify such predictions [12]; $\mathrm{Set}(b)$ is a nomological notion given M, not an epistemic one.

3 The logic of standpoints

Read $\Box A$ at b as "from every standpoint real from b, A", i.e. A holds at every $x \leq b$. The standpoint frame is (M, R) with bRx iff $x \leq b$. It is reflexive and transitive, so S4 is valid, and in causal spacetimes R is antisymmetric. Goldblatt proved that for the causal order of Minkowski spacetime the valid formulas are exactly the theorems of S4.2 [8]; by time reversal this applies to R.

The axiom .2 is $\Diamond\Box A \to \Box\Diamond A$. On reflexive transitive frames it is valid iff the frame is directed: whenever bRx and bRy there is z with xRz and yRz [3]. Applied to the subframe generated by b (the events $x \leq b$), this gives:

Proposition 8. *.2 is valid on the subframe generated by b if and only if any two events in $J^-(b)$ have a common causal past: $J^-(x) \cap J^-(y) \neq \emptyset$ for all $x, y \leq b$.*

Proof. Directedness of the generated subframe requires, for every $u \leq b$ and all $x, y \leq u$, some $z \leq x, y$; since $x, y \leq u \leq b$, this is the stated condition, and conversely the stated condition gives it for every u. □

In standpoint terms, .2 says that any two standpoints real from b share a common ground. Its failure means that b's fragment contains standpoints whose own fragments are disjoint: histories with no common origin, coordinated only by having b in their common future.

Proposition 9. *Let M be conformal to $(\eta_-, \infty) \times \mathbb{R}^3$ with $\eta_- > -\infty$, for instance a spatially flat FLRW universe with a big bang ($\eta_- = 0$), or one preceded by a finite inflationary era (more negative η_-). Then $x, y \in M$ have a common past iff $|X - Y| < (\eta_x - \eta_-) + (\eta_y - \eta_-)$. Hence .2 fails on the subframe generated by every event. If $\eta_- = -\infty$, .2 is valid everywhere.*

Proof. $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ for $c > \eta_-$. Two such balls meet for some $c > \eta_-$ iff $|X - Y| \leq (\eta_x - c) + (\eta_y - c)$ for some such c, i.e. iff $|X - Y| < \eta_x + \eta_y - 2\eta_-$, and a common point of the balls is a common past event. For any b, $J^-(b) \cap S_c$ is a ball of radius $\eta_b - c$, so for c close to η_- it contains points $x, y \in S_c$ with $|X - Y| = 2(\eta_b - c) > 2(c - \eta_-)$ once $c < (\eta_b + \eta_-)/2$; these have no common past. If $\eta_- = -\infty$ the bound is never violated. □

This is the horizon problem of standard cosmology in modal dress [9, 14]. Without inflation, two points of the last scattering surface in our past light cone, at comoving separation up to $2(\eta_0 - \eta_{\rm ls})$, have a common past only if $2(\eta_0 - \eta_{\rm ls}) < 2\eta_{\rm ls}$, which fails by a large factor: *the logic of our own standpoint is S4 but not S4.2.* An inflationary era lowers η_- enough that every pair of observed points has a common past; this restores .2 for the observed pairs, which is exactly what inflation was introduced to do. It does not restore .2 for all pairs in our past unless the inflationary era extends without past limit, which the Borde–Guth–Vilenkin theorem makes doubtful [1]. So even with inflation, our standpoint logic is S4 without .2, with .2 holding on a sub-domain large enough to contain everything we can observe.

Remark 10. Validity of S4 alone on all globally hyperbolic standpoint frames is immediate. Causal modal logics of general spacetimes are studied by Lewis and van der Schaaf [16], mostly for the future-directed relation, where confluence asks for a common *future*; the past-directed frame used here can behave differently (Proposition 9 has common futures for every pair). Whether S4 is exactly the logic of the class of all past-directed frames, and which logics arise from natural classes (compact Cauchy surfaces, black holes, causal sets), we leave open. We have not found the identification of the horizon problem with the failure of .2 in the sources read.

4 Where grounding matters

A standpoint-first ontology says that what is real is real from standpoints, and that the tenseless geometry is what the standpoints agree on. A block-first ontology says that the geometry of a single block is basic, and what is real from b is the restriction of the block to b's past. When every event is a standpoint, the two agree on every fact: every event lies in its own fragment, so the fragments cover M and their agreement fixes the geometry [5]. Suppose instead that the standpoints that exist are the events on matter worldlines, a natural restriction if a standpoint is a place where something happens. Let Γ be the union of those worldlines.

Definition 11. The *grounded region* is $G = \bigcup_{s\in\Gamma} J^-(s) = J^-(\Gamma)$, the events real from some existing standpoint. The *determined region* is $D = G \cup \mathrm{Set}(\Gamma)$, where $\mathrm{Set}(\Gamma)$ is the set of events all of whose past-inextendible causal curves meet $\bigcup_{s\in\Gamma} I^-_{\mathrm{s}}(s)$. The *shadow* is $M \setminus D$.

G is a past set, so no future-directed causal curve from outside G enters G, and data in G fix nothing to their past beyond G; deterministic laws extend G only by $\mathrm{Set}(\Gamma)$. On the shadow, the two ontologies disagree. The block-first ontology says the shadow has a geometry and contents, as definite as anything else. The standpoint-first ontology says nothing there is real from any standpoint, and nothing real from a standpoint fixes it; so, unless the laws carry more than domain-of-dependence information, there is no fact about it.

Proposition 12 (A lone observer in de Sitter)**.** *Let M be global de Sitter spacetime and Γ a single inextendible timelike geodesic. Then $D = J^-(\Gamma)$ up to its null boundary, and the shadow contains the nonempty open set $M \setminus J^-(\Gamma)$, which is the region beyond the observer's future event horizon.*

Proof. Conformally, M is $(-\pi/2, \pi/2) \times S^3$ and Γ is the curve $X = N$, a pole. Then $\bigcup_{s\in\Gamma} I^-_{\mathrm{s}}(s) = \{d(X, N) < \pi/2 - \eta\}$, the open region below the future event horizon [7]. For x outside it and $\eta_- < c < \eta_x$, $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ lies in $B(N, \pi/2 - c)$ iff $d(X, N) + \eta_x < \pi/2$ by Lemma 4, i.e. iff x is below the horizon. The proof of Theorem 2 applies with $\bigcup_s I^-_{\mathrm{s}}(s)$, an open past set, in place of $I^-_{\mathrm{s}}(b)$. So no event beyond the horizon is settled. □

Half of global de Sitter spacetime is thus real from no standpoint and not fixed by anything real from one. By contrast, in a homogeneous universe with matter everywhere, every event lies on or in the past of a matter worldline, the shadow is empty, and the two ontologies agree on every fact.

What this shows. The strongest objection to standpoint-first ontologies of time is that they agree with eternalism on everything and differ only in a claim about fundamentality, which may be idle. The shadow gives that claim a precise locus. The disagreement is real exactly when some region is in the past of no matter and is not fixed from such pasts by the laws. It is empty in the idealized homogeneous model of our universe, and large in sparse universes with event horizons. Earman's discussion of observational indistinguishability [5, 11] is the epistemic shadow of the same structure; the point here is that for a standpoint-first ontology the underdetermination is ontic, and its extent is computable.

Limits. Three caveats. The restriction of standpoints to matter is an extra premise; with every event a standpoint the shadow is empty. Laws that carry more than domain-of-dependence information, notably analyticity, would fix the shadow; Einstein's equations do not impose analyticity, and Manchak's results on observationally indistinguishable extensions [12] suggest that even the region we have called determined is fixed only given the spacetime, not from the grounded data alone. And test matter is an idealization: a real lone body in de Sitter would be a Schwarzschild–de Sitter spacetime, where the same horizon structure appears.

5 Conclusion

The companion theorem fixes the content of an event's fragment. Horizons then decide three further things: whether that fragment settles anything beyond itself, whether the standpoints in it share common ground, and whether a standpoint-first ontology differs from eternalism at all. The answers are computable from conformal structure. In our idealized universe they are: nothing beyond the past is settled; the standpoint logic is S4 without .2, with .2 restored by inflation on the observable domain; and the grounding dispute has an empty shadow.

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